

Design and Implementation of a Database Management System for Event Registration and Venue Scheduling System

User Interface Demo

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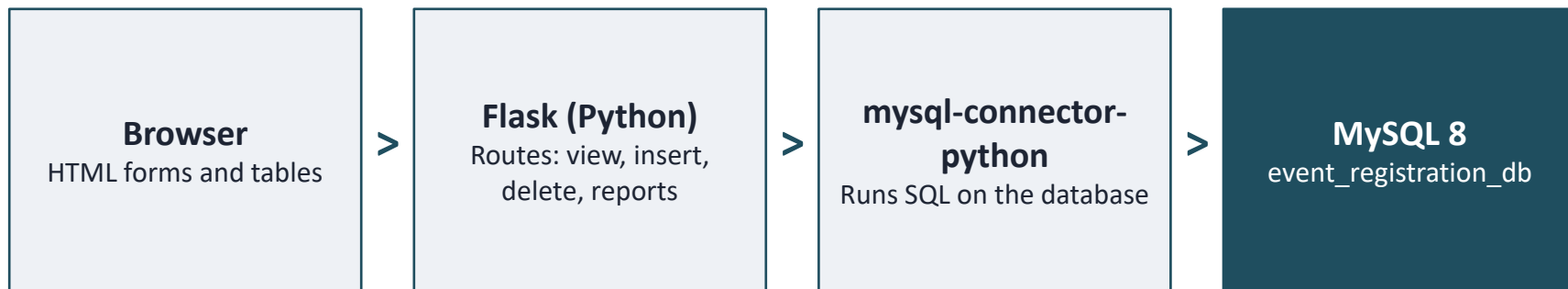
Where we left off

- 13 tables in 3NF covering the full event lifecycle: events, sessions, registrations, payments, attendance, certificates, feedback
- Business rules enforced as constraints: unique registration per participant and event, non-negative payments, venue slot uniqueness, valid ratings
- 10 queries and 4 views for reporting (seats, revenue, attendance, speaker schedule)

We built a web UI connected to the same MySQL database

Technology stack and database connectivity

Every action in the UI runs a real SQL statement



Constraint errors from MySQL are shown in the UI as they are returned.

View records

One page per table, with a live row count

Event Registration Console

participants

events

sessions

registrations

payments

attendance

certificates

feedback

venues

speakers

coordinators

categories

Reports (views)

registrations

Add a record, view all rows, or delete one.

participant_id

Aditya Kumar

event_id

TechFest 2026

registration_date

dd-mm-yyyy

status

confirmed

Add registration

9 row(s)

registration_id	participant_id	event_id	registration_date	status	
1	1	1	2026-08-25	confirmed	Delete
2	2	1	2026-08-25	confirmed	Delete
3	3	1	2026-08-26	confirmed	Delete
4	4	2	2026-08-27	confirmed	Delete
5	5	2	2026-08-27	confirmed	Delete
6	6	3	2026-08-28	confirmed	Delete
7	7	3	2026-08-28	confirmed	Delete
8	1	2	2026-08-29	confirmed	Delete
9	8	1	2026-08-30	cancelled	Delete

- Sidebar lists every table
- Rows are read with `SELECT *` from MySQL
- Same data as Workbench

Insert a record

Add a participant and confirm it in MySQL

participants
Add a record, view all rows, or delete one.

full_name

email

phone

Add participant

8 row(s)

participant_id	full_name	email	phone	
1	Aditya Kumar	aditya.kumar@mail.com	9800000001	Delete
2	Bhavna Roy	bhavna.roy@mail.com	9800000002	Delete
3	Chirag Patel	chirag.patel@mail.com	9800000003	Delete
4	Divya Menon	divya.menon@mail.com	9800000004	Delete
5	Esha Gupta	esha.gupta@mail.com	9800000005	Delete
6	Farhan Ali	farhan.ali@mail.com	9800000006	Delete
7	Gauri Joshi	gauri.joshi@mail.com	9800000007	Delete
8	Harsh Vardhan	harsh.vardhan@mail.com	9800000008	Delete

participants
Add a record, view all rows, or delete one.

full_name

email

phone

Add participant

9 row(s)

participant_id	full_name	email	phone	
1	Aditya Kumar	aditya.kumar@mail.com	9800000001	Delete
2	Bhavna Roy	bhavna.roy@mail.com	9800000002	Delete
3	Chirag Patel	chirag.patel@mail.com	9800000003	Delete
4	Divya Menon	divya.menon@mail.com	9800000004	Delete
5	Esha Gupta	esha.gupta@mail.com	9800000005	Delete
6	Farhan Ali	farhan.ali@mail.com	9800000006	Delete
7	Gauri Joshi	gauri.joshi@mail.com	9800000007	Delete
8	Harsh Vardhan	harsh.vardhan@mail.com	9800000008	Delete
9	Shrestha Mal	shrestha.official.005@gmail.com	06289078226	Delete

mysql> SELECT COUNT(*) FROM participants;
+-----+
| COUNT(*) |
+-----+
| 9 |
+-----+
1 row in set (0.009 sec)

Runs: `INSERT INTO participants (full_name, email, phone) VALUES (...)`

Delete a record

Remove the row and confirm the change in MySQL

participants

Add a record, view all rows, or delete one.

Add participant

9 row(s)

participant_id	full_name	email	phone	
1	Aditya Kumar	aditya.kumar@mail.com	9800000001	<button>Delete</button>
2	Bhavna Roy	bhavna.roy@mail.com	9800000002	<button>Delete</button>
3	Chirag Patel	chirag.patel@mail.com	9800000003	<button>Delete</button>
4	Divya Menon	divya.menon@mail.com	9800000004	<button>Delete</button>
5	Esha Gupta	esha.gupta@mail.com	9800000005	<button>Delete</button>
6	Farhan Ali	farhan.ali@mail.com	9800000006	<button>Delete</button>
7	Gauri Joshi	gauri.joshi@mail.com	9800000007	<button>Delete</button>
8	Harsh Vardhan	harsh.vardhan@mail.com	9800000008	<button>Delete</button>
9	Shrestha Mal	shrestha.offical.005@gmail.com	06289078226	<button>Delete</button>

Deleted 1 row from participants.

```
mysql> SELECT COUNT(*) FROM participants;
+-----+
| COUNT(*) |
+-----+
|      8 |
+-----+
1 row in set (0.009 sec)
```

participants

Add a record, view all rows, or delete one.

Add participant

8 row(s)

participant_id	full_name	email	phone	
1	Aditya Kumar	aditya.kumar@mail.com	9800000001	<button>Delete</button>
2	Bhavna Roy	bhavna.roy@mail.com	9800000002	<button>Delete</button>
3	Chirag Patel	chirag.patel@mail.com	9800000003	<button>Delete</button>
4	Divya Menon	divya.menon@mail.com	9800000004	<button>Delete</button>
5	Esha Gupta	esha.gupta@mail.com	9800000005	<button>Delete</button>
6	Farhan Ali	farhan.ali@mail.com	9800000006	<button>Delete</button>
7	Gauri Joshi	gauri.joshi@mail.com	9800000007	<button>Delete</button>
8	Harsh Vardhan	harsh.vardhan@mail.com	9800000008	<button>Delete</button>

Runs: DELETE FROM participants WHERE participant_id = ?

Business rules enforced by the database

Invalid data is rejected, not just hidden

Rejected by the database: Duplicate entry '1-1' for key 'registrations.uq_participant_event'

registrations

Add a record, view all rows, or delete one.

participant_id: Aditya Kumar | event_id: TechFest 2026 | registration_date: dd-mm-yyyy | status: confirmed | Add registration

9 row(s)

registration_id	participant_id	event_id	registration_date	status	
1	1	1	2026-08-25	confirmed	Delete
2	2	1	2026-08-25	confirmed	Delete
3	3	1	2026-08-26	confirmed	Delete
4	4	2	2026-08-27	confirmed	Delete
5	5	2	2026-08-27	confirmed	Delete
6	6	3	2026-08-28	confirmed	Delete
7	7	3	2026-08-28	confirmed	Delete
8	1	2	2026-08-29	confirmed	Delete
9	8	1	2026-08-30	cancelled	Delete

UNIQUE (participant_id, event_id)

Rejected by the database: Check constraint 'chk_amount_nonneg' is violated.

payments

Add a record, view all rows, or delete one.

registration_id: #1 Aditya Kumar - TechFest 2026 | amount: | payment_date: dd-mm-yyyy | mode: cash | Add payment

5 row(s)

payment_id	registration_id	amount	payment_date	mode	
1	4	500.00	2026-08-27	upi	Delete
2	5	500.00	2026-08-27	card	Delete
3	6	100.00	2026-08-28	cash	Delete
4	7	100.00	2026-08-28	upi	Delete
5	8	500.00	2026-08-29	upi	Delete

CHECK (amount >= 0)

Conclusion

Reports

Live results from the SQL views.

Seats booked per event

event_id	event_name	seats_booked
1	TechFest 2026	3
2	AI & Cloud Workshop	3
3	Cultural Night	2

Revenue per event

event_id	event_name	total_revenue
2	AI & Cloud Workshop	1500.00
3	Cultural Night	200.00
1	TechFest 2026	0.00

Attendance summary

registration_id	full_name	event_name	total_sessions	attended	attendance_pct
1	Aditya Kumar	TechFest 2026	3	2	66.7
2	Bhavna Roy	TechFest 2026	3	1	33.3
3	Chirag Patel	TechFest 2026	3	2	66.7
9	Harsh Vardhan	TechFest 2026	0	0	
4	Divya Menon	AI & Cloud Workshop	1	1	100.0
5	Esha Gupta	AI & Cloud Workshop	1	1	100.0
8	Aditya Kumar	AI & Cloud Workshop	1	1	100.0
7	Gauri Joshi	Cultural Night	1	1	100.0

Speaker schedule

speaker_id	speaker_name	event_name	venue_name	session_date	start_time	end_time
1	Dr. Ananya Rao	TechFest 2026	Auditorium A	2026-09-10	10:00:00	11:30:00
1	Dr. Ananya Rao	AI & Cloud Workshop	Seminar Hall B	2026-09-15	9:30:00	12:30:00
2	Mr. Karan Mehta	TechFest 2026	Auditorium A	2026-09-11	10:00:00	11:30:00
4	Ms. Priya Sen	Cultural Night	Open Air Theatre	2026-09-20	18:00:00	20:00:00
3	Prof. Leela Nair	TechFest 2026	Conference Room C	2026-09-12	14:00:00	15:30:00

Certificate eligible (attendance >= 60%)

registration_id	full_name	event_name	total_sessions	attended	attendance_pct
1	Aditya Kumar	TechFest 2026	3	2	66.7
3	Chirag Patel	TechFest 2026	3	2	66.7
4	Divya Menon	AI & Cloud Workshop	1	1	100.0
5	Esha Gupta	AI & Cloud Workshop	1	1	100.0
8	Aditya Kumar	AI & Cloud Workshop	1	1	100.0
6	Farhan Ali	Cultural Night	1	1	100.0
7	Gauri Joshi	Cultural Night	1	1	100.0

Conclusion

- Working UI performs insert, delete and view on MySQL
- Constraints protect data integrity

Future enhancements

- Login with roles for admin and participants
- Automatic venue overlap check and capacity check
- Certificate generation as PDF

Thank You!